Andy Ricardo Casafranca Carrascal

Mechatronics Engineering Graduate · Instrumentation, Control & Automation · Applied AI for Signals

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PROFILE

Mechatronics Engineering graduate (Bachelor's degree, UPC), ranked in the top 10% of my class. Field experience validating flowmeters with clamp-on ultrasonic meters at EMAPICA wells and reservoirs. Lead author of a paper under review at IEEE Access on detecting illegal water connections with machine learning. I combine instrumentation, Siemens S7-1200 PLCs, embedded systems and signal processing, with advanced English (C1). Available immediately, including mine-site rotations.

ENGINEERING EXPERIENCE

Instrumentation Intern — LabProjects · Ica, Peru — Sep 2026

- Validated flow at EMAPICA wells and reservoirs using clamp-on ultrasonic flowmeters as the reference.
- Adjusted the correction factors of installed meters to ensure accurate field data.

RESEARCH & PROJECTS

HydraDetect-AI — Detection of illegal drinking-water connections

- Portable system that detects illegal connections by analyzing pressure transients (water hammer) with machine learning: 89% field accuracy with offline on-device inference.
- Built the full pipeline: sensor acquisition, feature extraction (FFT, Welch PSD, wavelets), classification, and a Python desktop app for field operation.
- Lead author of the resulting paper, under review at IEEE Access.

EEG/EMG Signal Acquisition and Classification — Academic project, UPC — 2023 – 2024

- Designed the analog front-end of an EEG/EMG headset and its STM32 firmware for real-time capture.
- Developed a Python pipeline (NumPy, SciPy) reaching 87% accuracy in motor-imagery classification (8 channels).

ESP32 IoT Projects — Academic and personal projects — 2022 – 2023

- Built 3 end-to-end IoT projects (hardware, firmware and dashboards), including a smart-greenhouse mesh network and a water-quality monitor.
- Wrote ESP32 firmware in C/C++ with mesh networking and OTA updates, feeding React dashboards over WebSocket.

1st place, MINSA National Health Hackathon 2017 — video-game-based depression screening app

ADDITIONAL EXPERIENCE

Bilingual Agent (English/Spanish) — TP (Teleperformance) · Lima — Feb 2025 – Jun 2025

- Supported drivers of a U.S. last-mile delivery platform with orders, routes and payments via chat and phone, meeting strict response-time and quality targets.

Provider Service Representative — Teleperformance, Centene account · Lima — Feb 2023 – Aug 2023

- Advised U.S. healthcare providers in English on claims, eligibility and benefits (regulated environment).
- Built an internal web tool used by my team.

EDUCATION

Bachelor's Degree in Mechatronics Engineering — Universidad Peruana de Ciencias Aplicadas (UPC) — 2019 – 2026

- Top 10% of my class (Décimo Superior). Professional engineering title in progress.
- Relevant coursework: Control Systems, Digital Signal Processing, Microprocessors, Robotics, Machine Learning.

Interpretation and Translation (English) — ICPNA · Lima — 2019 – 2023

CERTIFICATIONS & COURSES

Applied Engineering for Mobile and Industrial Robotics — Universidad de Alcalá (Spain) · 10 h — Oct 2024

IBM Cybersecurity Analyst Specialization

Practical Ethical Hacking & Network Security — TCM Security / TryHackMe — 2024

SKILLS & LANGUAGES

- **Instrumentation & measurement:** clamp-on ultrasonic flowmeters, correction factors, analog front-end design.
- **Control & automation:** Siemens S7-1200 PLC (TIA Portal), PID, control systems, industrial and mobile robotics.
- **Embedded systems:** STM32, ESP32, Arduino, RTOS, I^2C, UART, PWM, PCB design.
- **Programming & data:** Python (NumPy, SciPy, PyTorch, TensorFlow, OpenCV), C/C++, MATLAB, Power BI, Excel.
- **Design & tools:** SolidWorks, Fusion 360, KiCad, Git, Linux, Docker.
- **Languages:** Spanish (native); English (advanced, C1).